

Pretom Saha

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EDUCATION

M.S. in Oceanography

Feb 2024 – May 2025

Shahjalal University of Science and Technology, Sylhet, Bangladesh

— CGPA: 3.70 / 4.00

B.Sc. (Honours) in Oceanography

Jan 2019 – Dec 2023

Shahjalal University of Science and Technology, Sylhet, Bangladesh

— CGPA: 3.52 / 4.00

RESEARCH INTERESTS

Geographic Information Systems (GIS), Remote Sensing, Natural Hazards and Disaster Resilience

RESEARCH EXPERIENCE

Undergraduate Thesis

Dynamics of Shoreline Changes on Bhasan Char Island, Bangladesh (2005–2023)

Supervisor: Abu Bokkar Siddique

- Extracted multi-temporal shorelines using **Google Earth Engine (JavaScript API)** from Landsat and Sentinel-2 imagery.
- Applied **Digital Shoreline Analysis System (DSAS)** to quantify erosion and accretion rates.

Tools: GEE, ArcGIS, DSAS **Keywords:** Shoreline Change, GIS, Remote Sensing

Course-Based Research Projects

- **Land Use and Land Cover Change Detection of Kutubdia Island, Bangladesh:** Performed supervised classification to generate LULC maps using RS–GIS techniques and assessed spatial patterns of land transformation driven by natural and anthropogenic factors.
- **Impact of Cyclone Sidr on Chlorophyll Distribution:** Analyzed Satellite data from NASA Ocean Color to quantify changes in chlorophyll-*a* concentration to assess cyclone-induced productivity changes in the Bay of Bengal.
- **Environmental Impact Assessment (EIA):** Evaluated environmental impacts of a cafeteria extension project at SUST and proposed mitigation strategies.
- **Seasonal Variability of Dissolved Oxygen:** Conducted spatial analysis using ArcGIS and ODV to interpret seasonal oxygen dynamics.

Research Assistant

Space and Environment Research Center (SERC), Bangladesh

- Conducted multi-temporal remote sensing analysis to monitor mangrove forest dynamics.
- Assisted in preprocessing, classification, and spatial interpretation of satellite data.

CONFERENCE PRESENTATION

International Conference on Oceanography (ICO 2024)

27–28 January 2024

Poster – “Dynamics of Shoreline Changes on Bhasan Char Island, Bangladesh: A Comprehensive Analysis from 2005 to 2023 Utilizing DSAS and GIS.”
Cox’s Bazar, Bangladesh

Organized by Bangladesh Oceanographic Research Institute (BORI).

[DOI](#)

WORKSHOPS & TRAINING

Workshop on “Remote Sensing for Mangrove Monitoring”

November 2024

Space and Environment Research Center (SERC), Bangladesh

- Focused on the application of remote sensing techniques for mangrove monitoring and ecosystem management.

TECHNICAL SKILLS

GIS and Remote Sensing Skills	– Satellite/Digital Image Analysis – Spatial Statistical Analysis
Programming Language	– Python, Google Earth Engine
RS/GIS Software Skills	– ESRI ArcGIS Pro, ArcMap – QGIS – ERDAS IMAGINE – ENVI
Oceanographic Tools	– Grab Sampler, Water Multi-parameter – Mini CTD – Plankton Net – Algal Torch

VOLUNTEERING & LEADERSHIP EXPERIENCE

Volunteer

24–25 March 2023

International Symposium on Marine Resource Management

Sylhet, Bangladesh

Held at Shahjalal University of Science and Technology (SUST).

REFERENCES

Abu Bokkar Siddique (Project Supervisor)

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Dr. Subrata Sarker (Head of Department)

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